9 Traceability

9.1 Extended traceability matrix

Requirement	Use case	Analysis classes	Component(s)	QAS / ADR	Evidence
FR-01	UC-05	StudentAccount	Authentication	QAS-03	Section 4
FR-02	UC-05	StudentAccount	Authentication	QAS-02, 03	Section 4
FR-03	UC-01	InternshipApplication	Application Management, Data Access	QAS-04	Section 5.2.1
FR-04	UC-01	InternshipApplication	Application Management	QAS-02	Section 5.2.1
FR-05	UC-01	InternshipApplication	Application Management	QAS-02	Section 5.2.1
FR-06	UC-02	InternshipApplication, StatusChange	Status Tracking		Section 5.2.2
FR-07	UC-02	StatusChange	Status Tracking		Section 5.2.2
FR-08	UC-01	InternshipApplication, Interview	Application Management		Section 5.2.1
FR-09	UC-03	Reminder	Reminder Management	ADR-03	Section 4
FR-10	UC-03	ProgressSummary	Dashboard	QAS-01	Section 4
FR-11	UC-04	InternshipApplication	Application Management		Section 4
FR-12	UC-01	Document	Document Management, File Storage Service	ADR-02	Section 5.2.1
QR-01	UC-03	ProgressSummary	Dashboard	QAS-01	Section 6.1
QR-02	UC-1,2, 3	StudentAccount	Authentication	QAS-02	Section 6.1
QR-03	UC-05	StudentAccount	Authentication	QAS-03	Section 6.1
QR-04	UC-01	InternshipApplication	Web UI, Application Management	QAS-04	Section 6.1
CON-01	UC-05	StudentAccount	Authentication	ADR-01	Section 2.4
CON-02	UC-1,2	InternshipApplication	Application Management	ADR-01	Section 2.4
CON-03	UC-03	Reminder	Reminder Management	ADR-03	Section 2.4
CON-04	All	—	Web UI	ADR-01	Section 2.4

9.2 Gap analysis

Gap	Items	What you did about it
Requirements with no component	None identified. All 20 requirements are assigned to at least one component in Section 9.1.	Reviewed the traceability matrix to confirm that each functional, quality, and constraint requirement has an assigned component.
Components with no requirement, QAS or ADR	None. All nine components are accounted for in Section 6.3.	Cross-checked the components against their associated requirements, use cases, QAS, and ADR references.
Use cases not analyzed (and why that is acceptable)	UC-03, UC-04	These use cases were included in the design but were not expanded in Section 4.2. UC-01 02, and 05 were prioritized as they covered core application management, status tracking, and authentication.